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The due date for submitting this assignment has passed.

Due on 2025-10-08, 23:59 IST.

Assignment submitted on 2025-10-05, 15:19 IST

Many repetitive online tasks can be automated using Python's **Selenium** library.
For example, logging into a website daily to check a dashboard can be automated with just a few steps.
First, install Selenium by running:

```
pip install selenium
```

Next, download the **Chrome WebDriver** that matches your version of Chrome.
To start writing the script, import the necessary modules:

```
from selenium import webdriver
from selenium.webdriver.common.by import By
from selenium.webdriver.common.keys import Keys
import time
```

Launch the Chrome browser:

```
driver = webdriver.Chrome()
```

Open the target website:

```
driver.get("https://www.sales-dashboard.com")
```

Suppose the login form has two input fields:

- Username with id="username"
- Password with id="password"

Enter the credentials:

```
driver.find_element(By.ID, "username").send_keys("user123")
driver.find_element(By.ID, "password").send_keys("mypassword")
```

Submit the form by simulating the **Enter** key:

```
driver.find_element(By.ID, "password").send_keys(Keys.ENTER)
```

Since the page may take time to load, pause execution briefly:

```
time.sleep(5)
```

Finally, close the browser:

```
driver.quit()
```

By following these steps, a Python script can automatically log into a website, eliminating the need for manual browser interaction.

1) Which Python library is used in the passage to automate browser tasks? *1 point*

- (•) Selenium
- ( ) BeautifulSoup
- ( ) Requests
- ( ) Pandas

Yes, the answer is correct.
Score: 1
Accepted Answers:
Selenium

2) Which command installs the Selenium library? *1 point*

- ( ) install selenium
- (•) pip install selenium
- ( ) python -m install selenium
- ( ) pip get selenium

Yes, the answer is correct.
Score: 1
Accepted Answers:
pip install selenium

3) What is the purpose of webdriver.Chrome() in the code? *1 point*

- ( ) To open a Firefox browser
- ( ) To install Chrome
- (•) To launch a Chrome browser instance
- ( ) To search Google

Yes, the answer is correct.
Score: 1
Accepted Answers:
To launch a Chrome browser instance

4) Which Selenium method opens a given URL in the browser? *1 point*

- ( ) open()
- ( ) driver.load()
- ( ) driver.goto()
- (•) driver.get()

Yes, the answer is correct.
Score: 1
Accepted Answers:
driver.get()

5) In the example, what is the id of the username input field? *1 point*

- ( ) name
- ( ) user
- (•) username
- ( ) user_id

Yes, the answer is correct.
Score: 1
Accepted Answers:
username

6) Which module provides the By class in Selenium? *1 point*

- (•) selenium.webdriver.common.by
- ( ) selenium.by
- ( ) webdriver.by
- ( ) selenium.common.by

Yes, the answer is correct.
Score: 1
Accepted Answers:
selenium.webdriver.common.by

7) What does send_keys(Keys.ENTER) simulate? *1 point*

- ( ) Typing text
- (•) Pressing the Enter key
- ( ) Clicking a button
- ( ) Submitting by mouse

Yes, the answer is correct.
Score: 1
Accepted Answers:
Pressing the Enter key

8) Why is time.sleep(5) used in the script? *1 point*

- (•) To delay closing the browser until the page loads
- ( ) To speed up execution
- ( ) To refresh the page
- ( ) To avoid importing libraries

Yes, the answer is correct.
Score: 1
Accepted Answers:
To delay closing the browser until the page loads

9) Which Selenium method closes the entire browser window? *1 point*

- ( ) driver.close()
- (•) driver.quit()
- ( ) browser.exit()
- ( ) webdriver.stop()

Yes, the answer is correct.
Score: 1
Accepted Answers:
driver.quit()

10) What is the main advantage of automating logins with Selenium? *1 point*

- (•) It saves manual effort
- ( ) It prevents internet outages
- ( ) It increases battery life
- ( ) It blocks ads

Yes, the answer is correct.
Score: 1
Accepted Answers:
It saves manual effort

Python provides powerful tools for handling dates, times, and calendar-related tasks through the **datetime** and **calendar** modules.

1. The datetime Module

The datetime module contains several classes, such as:

- **date** – Represents a calendar date (year, month, day).
- **time** – Represents a time of day (hours, minutes, seconds, microseconds).
- **datetime** – Combines both date and time into one object.
- **timedelta** – Represents the difference between two dates or times.

To get the current date and time:

```
import datetime
now = datetime.datetime.now()
```

his returns an object containing year, month, day, hour, minute, and second.
To create a specific date:

```
d = datetime.date(2025, 8, 10)
```

You can extract components:

```
now.year, now.month, now.day
```

You can also format dates using strftime:
Here:

- %A → Full weekday name
- %d → Day of month (zero-padded)
- %B → Full month name
- %Y → Year with century

2. The calendar Module

The calendar module is useful for working with entire months or years. Example: Printing a month's calendar:

```
import calendar
print(calendar.month(2025, 8))
```

Checking if a year is a leap year:

```
calendar.isleap(2024)  # Returns True
```

Getting the weekday of a given date:

```
calendar.weekday(2025, 8, 10)
```

11) Which function from datetime.date returns today's date? *1 point*

- ( ) date.current()
- ( ) date.now()
- (•) date.today()
- ( ) date.localtime()

Yes, the answer is correct.
Score: 1
Accepted Answers:
date.today()

12) What does %B stand for in date formatting? *1 point*

- ( ) Abbreviated month name
- ( ) Full weekday name
- (•) Full month name
- ( ) Day of the year

Yes, the answer is correct.
Score: 1
Accepted Answers:
Full month name

13) Which import statement allows direct use of date, time, and datetime without module prefix? *1 point*

- ( ) import datetime
- ( ) from datetime import module
- ( ) from datetime import all
- (•) from datetime import date, time, datetime

Yes, the answer is correct.
Score: 1
Accepted Answers:
from datetime import date, time, datetime

14) What type does calendar.isleap() return? *1 point*

- ( ) Integer
- ( ) String
- (•) Boolean
- ( ) Float

Yes, the answer is correct.
Score: 1
Accepted Answers:
Boolean

15) Which symbol in strftime gives a zero-padded day of the month? *1 point*

- ( ) %B
- ( ) %A
- (•) %d
- ( ) %m

Yes, the answer is correct.
Score: 1
Accepted Answers:
%d

16) If calendar.weekday(2023, 12, 31) returns 6, what day is that? *1 point*

- ( ) Saturday
- ( ) Monday
- ( ) Friday
- (•) Sunday

Yes, the answer is correct.
Score: 1
Accepted Answers:
Sunday

17) What will datetime.timedelta(days=10) create? *1 point*

- ( ) A future date 10 days later
- ( ) A calendar for 10 months
- ( ) A list of 10 dates
- (•) A time duration of 10 days

Yes, the answer is correct.
Score: 1
Accepted Answers:
A time duration of 10 days

18) Which strftime code outputs a 24-hour format hour? *1 point*

- ( ) %I
- (•) %H
- ( ) %p
- ( ) %h

Yes, the answer is correct.
Score: 1
Accepted Answers:
%H

19) What will datetime.date.today().weekday() return if today is Friday? *1 point*

- ( ) 6
- ( ) 1
- (•) 4
- ( ) 5

Yes, the answer is correct.
Score: 1
Accepted Answers:
4

20) Which module provides functions like isleap() and month()? *1 point*

- (•) calendar
- ( ) datetime
- ( ) time
- ( ) math

Yes, the answer is correct.
Score: 1
Accepted Answers:
calendar